

Qinan Wang

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EDUCATION

- **Johns Hopkins University** Baltimore, MD
PhD in Physics and Astronomy Sep 2017 - Aug 2023
- **University of Hong Kong** HKSAR, China
BSc in Mathematics/Physics (First Honor) Sep 2012 - June 2016

SKILLS SUMMARY

- **Program Languages:** Python, SQL
- **Data Reduction:** *TESS*, HST STIS/COS, APO 3.5m DIS/KOSMOS/ARCTIC, ATLAS

EXPERIENCE

- **Massachusetts Institute of Technology/Weizmann Institute of Science** Cambridge, MA, USA/Rehovot, Israel
TESS-ULTRASAT joint postdoc fellow Sep 2024 -
- **Johns Hopkins University** Baltimore, MD, USA
Postdoc Research Fellow - Dr. Armin Rest Sep 2023 - Sep 2024
- **University of Hong Kong** HKSAR, China
Research Assistant - Prof. Meng Su Aug 2016 - June 2017
 - Influence of instrumental contamination in CMB lensing reconstruction.
- **University of California, Berkeley** Berkeley, CA, USA
Research Assistant - Dr John Tomsick June 2015 - Aug 2015
 - Identification of X-ray counterparts of *INTEGRAL* sources in *Chandra*

HONORS AND AWARDS

- Rosita King Ho Scholarships, 2015
- Li Po Kwai Scholarships, 2014
- Alan John Ellis Prizes in Mathematics, 2013

GRANT & TELESCOPE TIME AND OBSERVATION EXPERIENCE

Accepted programs as Principle Investigator:

- *Hubble Space Telescope*, GO program in Cycle 33-35, 9 orbits. 2025 - 2028
“Phoenix from the Ashes: Searching for Surviving Companions to SN 2023bee with early excess”
- Magellan telescopes at the Las Campanas Observatory, 2 nights. 2025 - 2026
“Constructing a Robust Rubin Supernovae Sample at $0.1 < z < 0.4$ with Magellan”
- *Neil Gehrels Swift Observatory*, ToO program in Cycle 20, 50 triggers (75ks), \$40k. 2024
“Tracing the evolution of SNe Ibn/Icn with *swift*”
- The 3.5m ARC telescope at Apache Point Observatory, >20 half-nights, including 11 triggered ToO. 2019 - 2024
“Spectroscopic follow-up on extragalactic transients in *TESS* Field-of-View”

Accepted programs and Observation experience as Co-Is:

- *Hubble Space Telescope*, 9 proposals. 2019 - Now
- *James Webb Space Telescope*, 9 proposals. 2022 - Now
- Blanco 4-meter Telescope at CTIO, 5 nights. 2022 - Now

MENTORING AND TEACHING EXPERIENCE

- Mark Inffiah, MIT UROP, 1 publication in prep 2025
Constructing analytic model for *TESS* supernovae lightcurves with symbolic regression.
- Wendy Sun, MIT UROP 2025
Searching for quasi-periodic galactic nuclei transients in *TESS*.
- Sophie von Coelln, high school → CWRU undergraduate student, 1 publication in prep 2022 - 2025
Constraining the progenitor of SN 2023bee with nebular phase spectrum.
- Anika Goel, REU student at STScI, 1 publication 2022
- Kyle Dalrymple, JHU undergraduate student 2022 - 2023
- Sofia Rest, JHU undergraduate student, 1 publication 2021 - 2024

INVITED TALKS AND COLLOQUIMS

- Colloquium, Weizmann Institute of Science Nov 2025
- Transients From Space, Space Telescope Science Institute Mar 2025
- Monday Afternoon Talk, Massachusetts Institute of Technology April 2024
- Talk at the TESS Mission Update Meeting, Massachusetts Institute of Technology June 2023
- Remote talk, Tsinghua University May 2023
- TESS Science Talk, Massachusetts Institute of Technology Feb 2023

SELECTED TALKS AND PRESENTATIONS

- The 1st Science Meeting of the ET2.0 space mission, remote Aug 2025
- The Cosmic Lighthouse conference, Cambridge, UK June 2025
- Talk at AAS 245th, National Harbor Jan 2025
- The Progenitors of Supernovae and their Explosions conference, Dali, China Aug 2024
- The TESS Science Conference III, Massachusetts Institute of Technology July 2024
- Transients Down Under conference, Swinburne University of Technology Jan 2024
- Talk at AAS 243th, New Orleans Jan 2024
- Talk at CIERA Observers Group Meetings, Northwestern University Nov 2023
- Contributed seminar, Carnegie Mellon University Nov 2023
- SuperNova EXplosions (SNEX) Conference, Technion Aug 2023
- HotSci Talk series, Space Telescope Science Institute Aug 2023
- The Transient and Variable Universe conference, UIUC June 2023
- Boom! A Workshop on Explosive Transients with LSST, UIUC July 2022
- Contributed seminar, University of Melbourne June 2022

OPEN-SOURCE SOFTWARES

- *TESS* EXtragalactic Alert System (TEXAS). Github
- TESSREDUCE. Github
- YSE-PZ. Github
- ATCLEAN. Github

PROFESSIONAL SERVICES

- Organizer of the Extragalactic Transient parallel session at the *TESS* Science Conference III, 2024
- Peer-reviewer for ApJL, ApJ, 2023-present
- Panelist for NOIRLab programs, 2023
- Panelist for NASA funding programs, 2022

PUBLICATIONS

First-author publications:

1. **Qinan Wang**, David O. Jones, Justin DR Pierel, Matthew R. Siebert, WD'Arcy Kenworthy, Richard Kessler, Mi Dai et al. [arXiv:2512.25064]:
Feeling Blue: Constructing a Robust SALT3 UV Template and Constraining its Redshift Dependency
2. **Qinan Wang**, Anika Goel, Luc Dessart et al., MNRAS 530, no.2 (2024): 3906-3923, [arXiv:2305.05015]:
A Low-Mass Helium Star Progenitor Model for the Type Ibn SN 2020nxt
3. **Qinan Wang**, Armin Rest, Georgios Dimitriadis et al., ApJ 962, no. 2 (2024): 17, [arXiv:2305.03779]:
Flight of the Bumblebee: the Early Excess Flux of Type Ia Supernova 2023bee revealed by *TESS*, *Swift* and Young Supernova Experiment Observations
4. **Qinan Wang**, Patrick Armstrong, Yossef Zenati et al., ApJL 943, no. 2 (2023): L15 [arXiv:2211.03811]:
Revealing the Progenitor of SN 2021zby through Analysis of the *TESS* Shock-cooling Light Curve
5. **Qinan Wang**, Armin Rest, Yossef Zenati et al., ApJ 923, no. 2 (2021): 167 [arXiv:2108.13607]:
SN 2018agk: A Prototypical Type Ia Supernova with a Smooth Power-law Rise in Kepler (K2)

Co-author publications:

1. Zachary G. Lane, Ryan Ridden-Harper, Sofia Rest, Armin Rest, Conor L. Ransome, **Qinan Wang**, Clarinda Montilla et al. [arXiv:2511.15975].
SN 2019vxn: A Shocking Coincidence between Fermi and TESS.
2. Yize Dong, V. Ashley Villar, Anya Nugent, Griffin Hosseinzadeh, Ryan J. Foley, Christa Gall, Monica Gallegos-Garcia et al. [arXiv:2511.03926].
Spectral Diversity in Type Ibn Supernovae and the Large Host Offset of SN2024acyl.
3. T. Mera, C. Ashall, P. Hoefflich, K. Medler, M. Shahbandeh, C. R. Burns, E. Baron et al. [arXiv:2510.09600].
JWST Observations of SN 2024ggi II: NIRSpec Spectroscopy and CO Modeling at 285 and 385 Days Past the Explosion.
4. Yossef Zenati, **Qinan Wang**, Alexey Bobrick, Lindsay DeMarchi, Hila Glanz, Mor Rozner, Jacob E. Jencson et al. ApJ 992, no. 2 (2025): 9. [arXiv:2207.07146]:
SN 2019tsf: Evidence for Extended Hydrogen-poor CSM in the Three-peaked Light Curve of Stripped Envelope of a Type Ib Supernova
5. Geoffrey C. Clayton, R. Wesson, Ori D. Fox, Melissa Shahbandeh, Alexei V. Filippenko, Bryony Nickson, Michael Engesser et al. ApJ 991, no. 2 (2025): 133. [arXiv:2505.01574]
Very Late-Time JWST and Keck Spectra of the Oxygen-Rich Supernova 1995N
6. Rodrigo Angulo, Armin Rest, William P. Blair, Jacob Jencson, David A. Coulter, **Qinan Wang**, Ryan J. Foley et al. ApJS 280, no. 1 (2025): 29. [arXiv:2508.10155]
Optimizing Convolution Direction and Template Selection for Difference Image Analysis
7. I. A. Paniagua, W. B. Hoogendam, D. O. Jones, G. Dimitriadis, R. J. Foley, C. Gall, J. O'Brien et al. [arXiv:2508.13263]
The New Status Qvo? SN 2021qvo is Another 2003fg-like Type Ia Supernova with a Rising Light-Curve Bump

8. A. Gagliano, V. A. Villar, T. Matsumoto, D. O. Jones, C. L. Ransome, A. E. Nugent, D. Hiramatsu **et al.** ApJ 989, no. 2 (2025): 182 [arXiv:2502.19469]
Evidence for an Instability-Induced Binary Merger in the Double-Peaked, Helium-Rich Type II_n Supernova 2023zkd
9. K. Medler, C. Ashall, P. Hoefflich, E. Baron, J. M. DerKacy, M. Shahbandeh, T. Mera **et al.**, ApJ 993, no. 2 (2025): 191 [arXiv:2507.19727]
JWST observations of SN 2023ixf II: The panchromatic evolution between 250 and 720 days after the explosion
10. E. Baron, C. Ashall, J. M. DerKacy, P. Hoefflich, K. Medler, M. Shahbandeh, E. Fereidouni **et al.**, ApJ 994, no. 2 (2025): 249, [arXiv:2507.18753]
JWST Observations of SN 2024ggi I: Interpretation and Model Comparison of the Type II Supernova 2024ggi at 55 days Past Explosion
11. J. M. DerKacy, C. Ashall, E. Baron, K. Medler, T. Mera, P. Hoefflich, M. Shahbandeh **et al.** [arXiv:2507.18785]
JWST Observations of SN 2023ixf I: Completing the Early Multi-Wavelength Picture with Plateau-phase Spectroscopy
12. Erin E. Hayes, Suhail Dhawan, Kaisey S. Mandel, David O. Jones, Ryan J. Foley, Stephen Thorp, Matthew Grayling **et al.** MNRAS (2025): staf1056, [arXiv:2506.04309]:
Characterising the Standardisation Properties of Type Ia Supernovae in the z band with Hierarchical Bayesian Modelling
13. Tinyanont, Samaporn, Ori D. Fox, Melissa Shahbandeh, Tea Temim, Robert Williams, Kittipong Wangnok, Armin Rest **et al.** ApJ 985, no. 2 (2025): 198. [arXiv:2504.14009]
Large Cold Dust Reservoir Revealed in Transitional SN Ib 2014C by James Webb Space Telescope Mid-Infrared Spectroscopy
14. Szalai, Tamás, Szanna Zsíros, Jacob Jencson, Ori D. Fox, Melissa Shahbandeh, Arkaprabha Sarangi, Tea Temim **et al.** A&A 697 (2025): A132. [arXiv:2503.12950]
JWST/MIRI detects the dusty SN1993J about 30 years after explosion
15. Roxburgh, Hugh, Ryan Ridden-Harper, Andrew Moore, Clarinda Montilla, Brayden Leicester, Zachary G. Lane, James Freeburn **et al.** [arXiv:2502.16905]
TESSELLATE: Piecing Together the Variable Sky With TESS
16. Moriya, Takashi J., David A. Coulter, Christa DeCoursey, Justin DR Pierel, Kevin Hainline, Matthew R. Siebert, Armin Rest **et al.** PASJ (2025): psaf052, [arXiv:2501.08969]:
Properties of high-redshift Type II supernovae discovered by the JADES transient survey
17. Coulter, D. A., J. D. R. Pierel, C. DeCoursey, T. J. Moriya, M. R. Siebert, B. A. Joshi, M. Engesser **et al.** submitted to ApJ, [arXiv:2501.05513]:
Discovery of a likely Type II SN at $z=3.6$ with JWST
18. Shahbandeh, Melissa, Ori D. Fox, Tea Temim, Eli Dwek, Arkaprabha Sarangi, Nathan Smith, Luc Dessart **et al.** ApJ 985, no. 2 (2025): 262. [arXiv:2410.09142]:
JWST/MIRI Observations of Newly Formed Dust in the Cold, Dense Shell of the Type II_n SN 2005ip
19. Sebastian Gomez, Matt Nicholl, Edo Berger, Peter K. Blanchard, V. Ashley Villar, Sofia Rest, Griffin Hosseinzadeh **et al.** MNRAS 535, no. 1 (2024): 471-515,, [arXiv:2407.07946]:
The Type I Superluminous Supernova Catalog I: Light Curve Properties, Models, and Catalog Description
20. J. D. R. Pierel, M. Engesser, D. A. Coulter, C. Decoursey, M. R. Siebert, A. Rest, E. Egami **et al.** ApJL 971, no. 2 (2024): L32, [arXiv:2406.05089]:
Discovery of An Apparent Red, High-Velocity Type Ia Supernova at $z=2.9$ with JWST
21. M. R. Siebert, C. Decoursey, D. A. Coulter, M. Engesser, J. D. R. Pierel, A. Rest, E. Egami **et al.** ApJ 972, no. 1 (2024): L13. [arXiv:2406.05076]:
Discovery of a Relativistic Stripped Envelope Type Ic-BL Supernova at $z=2.83$ with JWST

22. Christa DeCoursey, Eiichi Egami, Justin DR Pierel, Fengwu Sun, Armin Rest, David A. Coulter, Michael Engesser **et al.** *ApJ* 979, no. 2 (2025): 250, [arXiv:2406.05060]:
The JADES Transient Survey: Discovery and Classification of Supernovae in the JADES Deep Field
23. S. Rest, A. Rest, C. D. Kilpatrick, J. E. Jencson, S. von Coelln, L. Strolger, S. Smartt **et al.** *ApJ* 979, no. 2 (2025): 114, [arXiv:2405.03747]:
ATClean: A Novel Method for Detecting Low-Luminosity Transients and Application to Pre-explosion Counterparts from SN 2023ixf
24. J. D. R. Pierel, A. B. Newman, S. Dhawan, M. Gu, B. A. Joshi, T. Li, S. Schuldt, **et al.** *ApJL* 967, no. 2 (2024): L37, [arXiv:2404.02139]:
Lensed Type Ia Supernova “Encore” at $z=2$: The First Instance of Two Multiply-Imaged Supernovae in the Same Host Galaxy
25. Szanna Zsíros, Tamás Szalai, Ilse De Looze, Arkaprabha Sarangi, Melissa Shahbandeh, Ori D. Fox, Tea Temim **et al.** *MNRAS* 529, no. 1 (2024): 155-168, [arXiv:2310.03448]:
Serendipitous detection of the dusty Type IIL SN 1980K with JWST/MIRI
26. S. Tinyanont, R. J. Foley, K. Taggart, K. W. Davis, N. LeBaron, J. E. Andrews, M. J. Bustamante-Rosell **et al.** *PASP* 136, no. 1 (2024): 014201, [arXiv:2309.07102]:
Keck Infrared Transient Survey I: Survey Description and Data Release 1
27. K. W. Davis, K. Taggart, S. Tinyanont, R. J. Foley, V. A. Villar, L. Izzo, C. R. Angus **et al.** *MNRAS* 523, no. 2 (2023): 2530-2550, [arXiv:2211.05134]:
SN 2022ann: a Type Icn supernova from a dwarf galaxy that reveals helium in its circumstellar environment
28. Melissa Shahbandeh, Arkaprabha Sarangi, Tea Temim, Tamas Szalai, Ori D. Fox, Samaporn Tinyanont, Eli Dwek **et al.** *MNRAS* 523, no. 4 (2023): 6048-6060, [arXiv:2301.10778]:
JWST observations of dust reservoirs in type IIP supernovae 2004et and 2017eaw
29. Hugh Roxburgh, Ryan Ridden-Harper, Zachary G. Lane, Armin Rest, Lancia Hubley, Rebekah Hounsell, **Qinan Wang et al.**, *ApJ* 963, no. 2 (2024): 89 [arXiv:2307.11294]:
A Comprehensive Investigation of Gamma-Ray Burst Afterglows Detected by *TESS*
30. D. A. Coulter, D. O. Jones, P. McGill, R. J. Foley, P. D. Aleo, M. J. Bustamante-Rosell, D. Chatterjee **et al.** *PASP* 135, no. 1048 (2023): 064501 [arXiv:2303.02154]:
YSE-PZ: A Transient Survey Management Platform that Empowers the Human-in-the-loop
31. Jacob E. Jencson, Jeniveve Pearson, Emma R. Beasor, Ryan M. Lau, Jennifer E. Andrews, K. Azalee Bostroem, Yize Dong **et al.** *ApJL* 952, no. 2 (2023): L30 [arXiv:2306.08678]:
A Luminous Red Supergiant and Dusty Long-period Variable Progenitor for SN 2023ixf
32. W. V. Jacobson-Galan, L. Dessart, R. Margutti, R. Chornock, R. J. Foley, C. D. Kilpatrick, D. O. Jones **et al.** accepted by *ApJL*, [arXiv:2306.04721]:
SN 2023ixf in Messier 101: Photo-ionization of Dense, Close-in Circumstellar Material in a Nearby Type II Supernova
33. P. D. Aleo, K. Malanchev, S. Sharief, D. O. Jones, G. Narayan, R. J. Foley, V. A. Villar **et al.** *ApJS* 266, no. 1 (2023): 9 [arXiv:2211.07128]:
The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae
34. M. D. Fulton, S. J. Smartt, L. Rhodes, M. E. Huber, V. A. Villar, T. Moore, S. Srivastav **et al.** *ApJL* 946, no. 1 (2023): L22 [arXiv:2301.11170]:
The optical light curve of GRB 221009A: the afterglow and the emerging supernova
35. Andreoni, Igor, Michael W. Coughlin, Daniel A. Perley, Yuhan Yao, Wenbin Lu, S. Bradley Cenko, Harsh Kumar **et al.** *Nature* 612, no. 7940 (2022): 430-434. [arXiv:2211.16530]:
A very luminous jet from the disruption of a star by a massive black hole

36. J. D. R. Pierel, D. O. Jones, W. D. Kenworthy, M. Dai, R. Kessler, C. Ashall, A. Do **et al.** ApJ 939, no. 1 (2022): 11 [arXiv:2209.05594]:
SALT3-NIR: Taking the Open-source Type Ia Supernova Model to Longer Wavelengths for Next-generation Cosmological Measurements
37. W. V. Jacobson-Galán, Padma Venkatraman, Raffaella Margutti, David Khatami, Giacomo Terreran, Ryan J. Foley, Rodrigo Angulo **et al.** ApJ 932, no. 1 (2022): 58 [arXiv:2203.03785]:
The Circumstellar Environments of Double-peaked, Calcium-strong Transients 2021gno and 2021inl
38. Samaporn Tinyanont, Ryan Ridden-Harper, R. J. Foley, Viktoriya Morozova, C. D. Kilpatrick, Georgios Dimitriadis, Lindsay DeMarchi **et al.** MNRAS 512, no. 2 (2022): 2777-2797 [arXiv:2110.10742]:
Progenitor and close-in circumstellar medium of type II supernova 2020fqv from high-cadence photometry and ultra-rapid UV spectroscopy
39. Ori D. Fox, Schuyler D. Van Dyk, Benjamin F. Williams, Maria Drout, Emmanouil Zapartas, Nathan Smith, Dan Milisavljevic **et al.** ApJL 929, no. 1 (2022):L15 [arXiv:2203.01357]:
The Candidate Progenitor Companion Star of the Type Ib/c SN 2013ge
40. W. V. Jacobson-Galán, Luc Dessart, D. O. Jones, Raffaella Margutti, D. L. Coppejans, Georgios Dimitriadis, Ryan J. Foley **et al.** ApJ 924, no. 1 (2022):15 [arXiv:2109.12136]:
Final Moments. I. Precursor Emission, Envelope Inflation, and Enhanced Mass Loss Preceding the Luminous Type II Supernova 2020tlf
41. Ryan Ridden-Harper, Armin Rest, Rebekah Hounsell, Tomás E Müller-Bravo, **Qinan Wang**, Villar, V. A.. arXiv preprint [arXiv:2111.15006]:
TESSreduce: transient focused *TESS* data reduction pipeline
42. D. O. Jones, R. J. Foley, G. Narayan, Jens Hjorth, M. E. Huber, P. D. Aleo, K. D. Alexander **et al.** ApJ 908(2), 143 (2021) [arXiv:2010.09724]:
The Young Supernova Experiment: survey goals, overview, and operations
43. Jeffrey Iuliano, Joseph Eimer, Lucas Parker, Gary Rhoades, Aamir Ali, John W. Appel, Charles Bennett **et al.** Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy IX, vol. 10708, pp. 259-277. SPIE, 2018. [arXiv:1807.04167]:
The cosmology large angular scale surveyor receiver design.
44. Kathleen Harrington, Joseph Eimer, David T. Chuss, Matthew Petroff, Joseph Cleary, Martin DeGeorge, Theodore W. Grunberg **et al.** Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy IX, vol. 10708, pp. 369-390. SPIE, 2018. [arXiv:1807.03807]:
Variable-delay polarization modulators for the CLASS telescopes.
45. Sumit Dahal, Aamir Ali, John W. Appel, Thomas Essinger-Hileman, Charles Bennett, Michael Brewer, Ricardo Bustos **et al.** Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy IX, vol. 10708, pp. 230-245. SPIE, 2018. [arXiv:1807.03927]:
Design and characterization of the cosmology large angular scale surveyor (CLASS) 93 GHz focal plane.
46. H. F. Chau, Cardythy Wong, **Qinan Wang**, Tieqiao Huang. arXiv preprint [arXiv:1608.08329]:
Qudit-Based Measurement-Device-Independent Quantum Key Distribution Using Linear Optics
47. H. F. Chau, **Qinan Wang**, Cardythy Wong. PRA 95, no. 2 (2017): 022311. [arXiv:1603.02370]:
Experimentally feasible quantum-key-distribution scheme using qubit-like qudits and its comparison with existing qubit-and qudit-based protocols
48. John A. Tomsick, Roman Krivonos, **Qinan Wang** et al., ApJ 816, no. 1 (2015): 38 [arXiv:1512.00044]:
Chandra observations of eight sources discovered by *INTEGRAL*